

Practice Quiz: Agents

Part A: Multiple Choice

1. Interoception is the perception of: A) External visual stimuli B) Internal bodily states such as heartbeat, breathing, and gut sensations C) Other people's emotional states D) The spatial layout of the environment
2. The anterior insular cortex is primarily associated with: A) Visual perception and object recognition B) Interoceptive processing and integration of internal bodily signals C) Motor planning and execution D) Language production (Broca's area)
3. The rubber hand illusion demonstrates that: A) Visual perception always overrides tactile perception B) Body ownership is an inference based on multisensory prediction error minimization C) The brain cannot distinguish between real and artificial limbs D) Touch is the most reliable sensory modality
4. In Active Inference, depersonalization may result from: A) Excessive precision on interoceptive prediction errors B) Low precision on interoceptive prediction errors, attenuating the self-model C) Damage to the motor cortex D) Overactivity in the visual cortex
5. Body integrity identity disorder (BIID) illustrates: A) A mismatch between the physical body and the generative model's body representation B) A purely psychiatric condition with no neurological basis C) Normal variation in body image D) A failure of motor control
6. Emotional experience, according to interoceptive inference accounts, is: A) A purely cognitive judgment about a situation B) A prediction about the causes of interoceptive signals (a "gut feeling" is literally a prediction) C) An entirely separate system from perception and cognition D) A behavioral response with no subjective component
7. The somatic marker hypothesis (Damasio) proposes that: A) The body has no role in decision-making B) Bodily signals (heart rate, gut feelings) function as emotional tags

that guide decisions C) Somatic markers are always accurate D) All emotional processing occurs in the amygdala

Part B: Short Answer

1. Explain how the rubber hand illusion could be disrupted if the experimenter introduced asynchronous stroking (200ms delay). What does this predict about the temporal precision requirements of multisensory integration?
2. Patient describes feeling "detached" from his body after trauma. Suggest a therapeutic approach based on Active Inference (increasing interoceptive precision) and explain why it might help.
3. Compare phantom limb syndrome (present model, missing limb) with body integrity identity disorder (missing model component, present limb). How are they inverse disruptions of the same self-modeling process?